Probability Recap

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Random Variable

Intuition

- Take on some random values
- Associate a distribution to the possible values

Example

Let X be the outcome of a die roll.

Possible values?

Possible distribution?

Expected Value

Intuition

Average over all possible outcome of X

$$\mathbb{E}[X] = \sum_{x \in X} x \Pr[X = x]$$

Example

Let X be the outcome of a fair dice roll. Then the expected value is $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5.$

Variance

Intuition

Quantify how much deviation from expected value.

$$\mathrm{Var}[x] = \mathbb{E}[(x - \mathbb{E}[x])^2]$$

Example

Let x be the outcome of a fair dice roll. Then the variance is $\mathrm{Var}[x] = \frac{1}{6}\left((1-3.5)^2 + \cdots + (6-3.5)^2\right) = \frac{35}{12}$.

Point Estimate – Intuition

Single best guess of some quantity of interest

Dependent on data

Point Estimate – Example

Suppose we want to know the mean income of a population However it is unlikely that we can collect data from everyone in the population. Instead, we ask a subset of the population and use the sample mean.

What is the quantity of interest?

What is the single best guess?

Point Estimate – Example

Example

Suppose we want to predict the housing price y from the size of the house x, and that the relationship is governed by:

$$y = \theta_0 + \theta_1 x.$$

Here, θ_0 and θ_1 are the quantities of interest. After we collect some data, we can use linear regression to generate a point estimate for θ_0 and θ_1.

Point Estimate – Warning

Point estimate depends on data collected

Data collected is random

Therefore the point estimate is a random variable

Makes sense to talk about the expected value and variance of the point estimate

Bias of Point Estimates – Intuition

The bias measures, on average, how much does the point estimate differs from the true value of the parameter.

Let $\hat{\theta}$ be the point estimate for θ. Then we define

$$\mathrm{Bias}\left[\hat{\theta}\right] = \mathbb{E}\left[\hat{\theta}\right] - \theta$$

Bias of Point Estimates – Example

Suppose we want to estimate the population mean μ. We do so by taking the average of m data $X^{(1)}, \ldots, X^{(m)}$. Thus, our point estimate for the actual μ is:

$$\hat{\mu} = \frac{1}{m} \sum_{i=1}^{m} X^{(i)}$$

Bias of Point Estimates – Example

The bias of this point estimate is $\mathrm{Bias}[\hat{\mu}] = \mathbb{E}[\hat{\mu}] - \mu$

$$\mathbb{E}[\hat{\mu}] - \mu = \left(\frac{1}{m} \sum_{i=1}^{m} \mathbb{E}[X^{(i)}] \right) - \mu$$

Since $\mathbb{E}\left[X^{(i)}\right] = \mu$, the summation term is μ and the bias is 0.

Variance of Point Estimates – Intuition

A point estimate varies according to your data.

The variance of the point estimate $\mathrm{Var}[\hat{\theta}]$ quantifies that.

Variance of Point Estimates – Example

Let us try to find the variance of the point estimate $\hat{\mu}$ from the previous example.

We shall assume that the population variance is σ^2. Then:

$$\mathrm{Var}[\hat{\mu}] = \mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m} X^{(i)}\right) = \frac{1}{\mathrm{m}^2}\left(\sum_{i=1}^{m} \mathrm{Var}\left[X^{(i)}\right]\right) = \frac{\sigma^2}{m}$$

Resources

MIT 18.05 - Introduction to Probability and Statistics

https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/

All of Statistics (Larry Wasserman)

Chapter 2 to Chapter 6 for probability recap

Chapter 7 for point estimation

The Learning Problem

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Terminology

Target Function

True underlying function we hope to estimate.

Hypothesis Space

The set of functions that we can pick from as an estimate to the target function.

Learning Algorithm

The algorithm that decides which function from the hypothesis space to use.

Math Notation

$\mathcal{H}$	The hypothesis space of a particular learner A.
f	The target function, which might not necessarily be in $\mathcal{H}$.
h	The function selected by the learning algorithm A. Note that $h \in \mathcal{H}$. Also denoted as $\hat{f}$, where the hat ($\hat{}$) denotes approximation.
$\mathcal{D}$	The set of observed data.
A	The learning algorithm that takes in $\mathcal{D}$ and outputs h.
$\mathcal{A}$	The set of all learning algorithms. We pick $A \in \mathcal{A}$. Each A has a particular inductive learning bias.

Example

Suppose we collect the data:

$$\mathcal{D} = \{(0,0), (0.04, 0.04), (0.12, 0.12)\}$$

generated by some f. Let our hypothesis space $\mathcal{H} = \{h_1, h_2\}$, with $h_1(x) = \sin x$ and $h_2(x) = \cos x$.

Then, our learning algorithm A picks an $h \in \mathcal{H}$ that best describes $\mathcal{D}$.

An example of A is the use of linear regression with gradient descent to minimize the mean square error.

Caveats

Warning 1

$\mathcal{H}$ might not include the target function f.

Warning 2

h is a random variable, dependent on the set of data $\mathcal{D}$ it gets to learn from (also assumed to be randomly sampled).